

MemDistill: Diagnosing Why Distilling RL Agent Memory Fails

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Abstract

Reinforcement learning (RL) can train language model agents to coordinate across multiple objectives simultaneously, but distilling these policies into smaller models via supervised fine-tuning (SFT) fails catastrophically. We present a diagnostic framework that localizes this failure across three levels of analysis, demonstrated on MEM1, a 7B RL agent for multi-objective retrieval-augmented QA. First, we establish that per-objective Exact Match degrades monotonically with objective count ($0.62 \rightarrow 0.341$ from $N = 1$ to 16), setting “slow the slope” as the distillation target. Second, SVD analysis reveals that RL weight changes concentrate $2\text{--}6\times$ supra-random along the base model’s principal directions, while SFT recovers significantly less of this geometric structure (alignment ratio $1.47\times$ vs. $2.78\times$). Third, matched-sample DPO—a stronger signal than SFT—improves over SFT by only $+1.0$ percentage points at $N=8$ ($p = 0.163$, not significant), and we show that an earlier apparent advantage was partly a truncation confound. A causal intervention—providing oracle retrieval at inference—recovers `em_partial` from 0.025 to 0.498 ($+47.3\text{pp}$, $p < 0.0001$), and even standard BM25 retrieval recovers $+12.2\text{pp}$, confirming that the student learns a viable search strategy but cannot execute it without the retrieval backend. A 7B capacity-isolation experiment quantifies the model-size contribution at *at most* $\approx 12\text{--}14\%$ of the teacher-student gap, remarkably stable across all objective counts. Our findings, drawn from a single MEM1-style multi-objective RL policy, redirect attention from training-signal richness toward the structural incompatibility between RL procedural policies and single-pass distillation.

1 Introduction

Reinforcement learning (RL) can endow language models with multi-objective capabilities that supervised fine-tuning alone does not produce. For instance, the MEM1 agent (Zhou et al., 2025)—a Qwen2.5-7B model trained with PPO via `veRL` (Sheng et al., 2024)—sustains a per-objective Exact Match (EM) of 0.62 when answering a single question, and still achieves 0.455 at $N = 8$ simultaneous objectives in a retrieval-augmented QA task. The same base model, without RL, collapses to $\text{EM} \leq 0.15$ for $N \geq 2$.

Deploying a 7B RL policy as an always-on sidecar is expensive. A natural solution is to distill the policy into a smaller ($<3\text{B}$) student via SFT on teacher trajectories—the standard recipe for knowledge distillation in LLMs (Hsieh et al., 2023; Gu et al., 2024). Yet when we apply this recipe to MEM1’s multi-objective trajectories, the result is catastrophic: a Qwen2.5-3B student trained on 1,118 RL trajectories achieves `em_partial` = 0.025 at $N = 8$ —a 94.5% relative drop from the teacher—with 83% of outputs truncated before the model finishes generating. While some degradation is expected from the $7\text{B} \rightarrow 3\text{B}$ capacity reduction, and though we lack a 3B RL baseline to precisely quantify the capacity contribution, the magnitude of degradation (from 0.455 to 0.025) strongly suggests factors beyond model size.

Why does distillation fail so completely for multi-objective RL policies? Prior work on RL-to-SFT gaps has examined alignment tax (Askell et al., 2021) and mode collapse in single-objective settings (Jin et al., 2025a), but the multi-objective regime—where the teacher must coordinate across N parallel goals within a single generation—remains unexplored.

We present MEMDISTILL, a three-phase di-

agnostic framework that progressively localizes the distillation failure from task-level symptoms to representation-level mechanisms to behavior-level explanations (Figure 1):

1. **Collapse curve** (§2.1): We establish that multi-objective EM degrades *monotonically* with objective count N , providing no evidence for a phase transition within the tested range ($N \leq 16$) and setting “slow the slope” as the distillation target.
2. **SVD alignment analysis** (§2.2): We show that RL weight changes concentrate $2\text{--}6\times$ supra-random along the base model’s principal directions (depending on subspace rank k ; see Table 1), while SFT distillation recovers less of this geometric signature (alignment ratio $1.47\times$ vs. $2.78\times$ for RL). The geometric gap raises the question of whether richer training signals can recover RL-like weight structure, motivating the next phase.
3. **Matched-sample distillation** (§3): DPO does not close the gap ($+1.0\text{pp}$, $p = 0.163$). A causal intervention—providing oracle retrieval at inference—recovers `em_partial` from 0.025 to 0.498 at $N = 8$ ($+47.3\text{pp}$, $p < 0.0001$), confirming that the student learns a viable search strategy but cannot execute it without the retrieval backend.

Contributions.

- A **diagnostic methodology** combining collapse characterization, weight-space geometric analysis (SVD alignment ratio), and matched-sample preference learning, with a truncation confound detection protocol for verbose RL-distilled outputs.
- An **empirical finding** that RL weight changes concentrate $2\text{--}6\times$ supra-random along the base model’s principal directions, while SFT does not recover this signature.
- A **negative result with causal explanation**: DPO fails to close the gap ($+1.0\text{pp}$, $p = 0.163$). Providing oracle retrieval at inference recovers `em_partial` from 0.025 to 0.498 at $N = 8$ ($p < 0.0001$), causally supporting behavioral mismatch: the student learns a search strategy it cannot execute.

2 Diagnostic Framework

We develop a three-phase diagnostic framework that progressively localizes the distillation failure from task-level symptoms to representation-

level mechanisms. Phase 1a quantifies how per-objective performance degrades as objective count N increases, establishing the shape of the collapse. Phase 1b uses singular value decomposition (SVD) to compare the geometric footprint of RL and SFT weight changes in the base model’s principal subspace. Phase 1c tests whether activation-level probes can recover per-objective performance signals, yielding a negative result that bounds the mechanism to weight-space geometry rather than activation-space encoding.

2.1 Phase 1a: Collapse Curve

We first ask: *does multi-objective EM degrade gradually or exhibit a sharp phase transition?* The answer determines whether distillation should target a cliff recovery or a slope reduction.

Setup. We evaluate the MEM1 RL policy (Qwen2.5-7B + PPO, trained via `veRL` (Sheng et al., 2024)) and the Qwen2.5-7B base model on a multi-objective RAG QA task drawn from FlashRAG (Jin et al., 2025b). Each prompt contains N questions separated by semicolons ($N \in \{1, 2, 4, 8, 12, 16\}$). The agent operates iteratively—`think` $\rightarrow$ `search` $\rightarrow$ `information` $\rightarrow$ `answer`—and receives a normalized Exact Match (EM) score per objective (SQuAD standard). We report `em_partial`: the mean per-objective EM averaged across all objectives and prompts at each N .

Results. Figure 2 plots `em_partial` for both models across N . MEM1’s performance degrades monotonically from 0.62 at $N = 1$ to 0.341 at $N = 16$ (a 45% relative drop). A linear model provides a reasonable first-order approximation ($\text{EM} \approx 0.62 - 0.018 \cdot N$, $R^2 = 0.984$; largest residual -2.2pp at $N = 8$), and fits better than a $\log N$ model ($R^2 = 0.949$). We use this as a descriptive summary, not a mechanistic claim about the form of the degradation. Holm–Bonferroni corrected pairwise comparisons against $N = 2$ reach significance at $N = 12$ and $N = 16$ ($p < 0.05$); the overall effect size is large (Cohen’s $d = 0.97$). No inflection point or S-curve is observed.

The base model, by contrast, collapses immediately. At $N = 1$ its EM is already 0.30 (half of MEM1), and for $N \geq 2$ it fluctuates between 0.075 and 0.148 with frequent answer-extraction failures (the base model often fails to produce

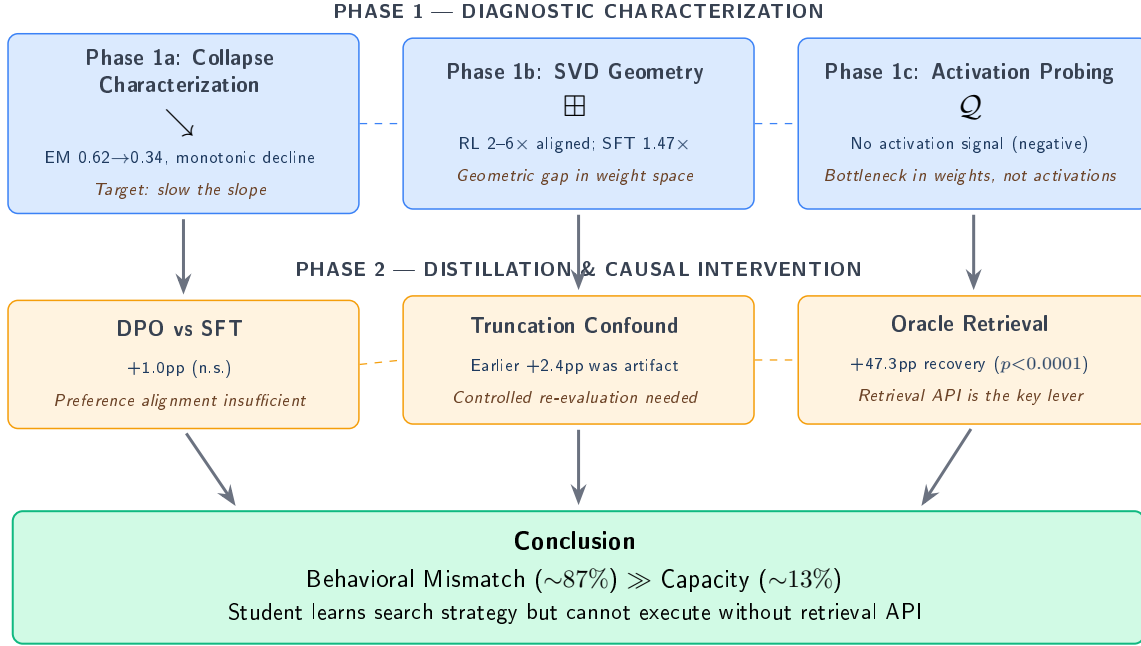


Figure 1: Overview of the MEMDISTILL diagnostic framework. Phase 1 (top) progressively characterizes the distillation failure through collapse analysis, SVD weight geometry, and activation probing. Phase 2 (middle) tests interventions: DPO preference learning, truncation confound correction, and oracle retrieval. The combined evidence (bottom) attributes ~87% of the teacher–student gap to behavioral mismatch.

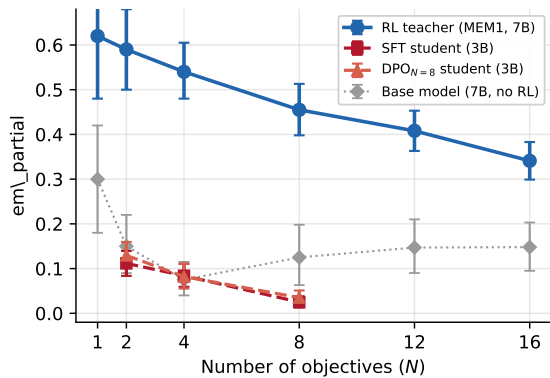


Figure 2: Multi-objective performance collapse. The RL teacher (MEM1) degrades monotonically with objective count N , while both SFT and DPO students retain <8% of the teacher’s capability at $N=8$. The base model (no RL) collapses immediately. Error bars show 95% bootstrap CIs (10,000 resamples).

183 text parseable as a structured multi-objective an-
184 swer). The RL–base gap is evident from $N=1$
185 and widens with N : at $N=8$, MEM1 achieves
186 0.455 vs. the base’s 0.125.

187 **Implication.** The monotonic degradation pro-
188 file has a direct consequence for distillation: the
189 target is to *slow the decline*, not to recover per-
190 formance after a phase transition. Any student
191 model that merely preserves base-level capabil-
192 ity will fail across the board; the distillation must
193 transfer the RL-specific mechanism that sustains
194 the shallow slope. This motivates the next phase:
195 identifying what that mechanism looks like in
196 weight space.

197 2.2 Phase 1b: SVD Alignment Analysis

198 Phase 1a establishes that RL training endows
199 a multi-objective capability absent in the base
200 model. We now ask: *where does the RL fine-*
201 *tuning place its weight changes relative to the*
202 *base model’s existing structure, and does SFT*
203 *distillation land in the same region?*

204 **Method.** Let $\mathbf{W}_{\text{base}} \in \mathbb{R}^{m \times n}$ denote a weight
205 matrix from the base model and $\mathbf{W}_{\text{ft}}$ the corre-
206 sponding matrix from a fine-tuned model (RL
207 or SFT). We compute the weight delta $\Delta \mathbf{W} =$
208 $\mathbf{W}_{\text{ft}} - \mathbf{W}_{\text{base}}$ and measure how much of $\Delta \mathbf{W}$
209 falls within the base model’s principal subspace.

Concretely, we compute the truncated SVD of

$\mathbf{W}_{\text{base}}$:

$$\mathbf{W}_{\text{base}} = \mathbf{U}\Sigma\mathbf{V}^\top \quad (1)$$

and retain the top- k left singular vectors $\mathbf{U}_k \in \mathbb{R}^{m \times k}$ and right singular vectors $\mathbf{V}_k \in \mathbb{R}^{n \times k}$. The *observed alignment* of $\Delta\mathbf{W}$ with this subspace is the fraction of its Frobenius norm captured by the projection:

$$\alpha_{\text{obs}}(k) = \frac{\|\mathbf{U}_k^\top \Delta\mathbf{W} \mathbf{V}_k\|_F}{\|\Delta\mathbf{W}\|_F} \quad (2)$$

In high-dimensional spaces, even random matrices exhibit non-trivial projection onto any fixed subspace due to concentration of measure. To calibrate, we generate 100 random matrices of the same dimensions as $\Delta\mathbf{W}$ (entries drawn i.i.d. from $\mathcal{N}(0, \sigma^2)$, where σ^2 matches the variance of $\Delta\mathbf{W}$) and compute their mean alignment $\alpha_{\text{rand}}(k)$. The *alignment ratio* is:

$$r(k) = \frac{\alpha_{\text{obs}}(k)}{\alpha_{\text{rand}}(k)} \quad (3)$$

A ratio of $r = 1$ indicates alignment indistinguishable from random; $r > 1$ indicates supra-random concentration along the base model’s principal directions. We apply this analysis to all 196 weight matrices in the 28-layer Qwen2.5-7B architecture.

We also decompose the projection into *U-side* and *V-side* components. The U-side alignment $\|\mathbf{U}_k^\top \Delta\mathbf{W}\|_F / \|\Delta\mathbf{W}\|_F$ measures how much $\Delta\mathbf{W}$ concentrates along the base model’s output directions (left singular vectors), while the V-side alignment $\|\Delta\mathbf{W} \mathbf{V}_k\|_F / \|\Delta\mathbf{W}\|_F$ captures concentration along input directions (right singular vectors). This decomposition reveals whether RL modifies how a layer *reads* its input or *writes* its output.

RL weight changes are supra-random aligned with the base subspace. Table 1 reports the alignment ratio $r(k)$ for the RL-base delta ($\Delta\mathbf{W}_{\text{RL}} = \mathbf{W}_{\text{RL}} - \mathbf{W}_{\text{base}}$) across $k \in \{64, 128, 256, 512, 1024\}$. At $k = 64$, the ratio is $6.48\times$ —RL weight changes concentrate along the base model’s top 64 singular directions at nearly $6.5\times$ the rate expected by chance. The ratio decreases monotonically as k grows ($4.11\times$ at $k = 128$, $2.78\times$ at $k = 256$, $2.01\times$ at $k = 512$, $1.64\times$ at $k = 1024$), consistent with changes that are tightly focused on the most prominent base directions rather than uniformly distributed.

	$k=64$	$k=128$	$k=256$	$k=512$	$k=1024$
RL-base	6.48	4.11	2.78	2.01	1.64
SFT-base	2.92	—	1.47	—	—
RL-SFT	3.80	—	1.81	—	—
<i>RL/SFT ratio at matched k:</i>					
$r_{\text{RL}}/r_{\text{SFT}}$	2.22	—	1.89	—	—

Table 1: SVD alignment ratio $r(k)$ (Eq. 3) for three weight deltas, averaged over 196 matrices. RL-base alignment is $2\text{--}6\times$ supra-random and consistently stronger than SFT-base (Wilcoxon $p < 1.3 \times 10^{-32}$ at all k ; per-layer distributions in Figure 3). Entries marked “—” indicate k values not computed for SFT/RL-SFT deltas due to computational budget; the reported $k \in \{64, 256\}$ span the range.

This finding contradicts the intuition that RL “expands” the model into novel directions. Instead, RL fine-tuning operates by *refining* the base model’s existing representational structure—adjusting weights preferentially along directions that already carry the most variance.

SFT distillation lands in a different subspace.

To enable same-architecture comparison, we train a full-rank SFT model (Qwen2.5-7B) on RL teacher trajectories and apply the same analysis. Using the same architecture and training regime (full-rank fine-tuning for both RL and SFT) ensures that alignment ratios reflect differences in *training objective* (PPO vs. behavior cloning) rather than dimensional or rank confounds. The distillation experiments in §3 use a separate Qwen2.5-3B student (LoRA rank 32), reflecting the deployment cost constraint. The SFT-base alignment ratio is consistently lower than RL-base: $2.92\times$ vs. $6.48\times$ at $k = 64$ and $1.47\times$ vs. $2.78\times$ at $k = 256$ (RL/SFT ratio $\approx 2 : 1$ at both k values). The RL-SFT delta ($\Delta\mathbf{W}_{\text{RL-SFT}} = \mathbf{W}_{\text{RL}} - \mathbf{W}_{\text{SFT}}$) has an alignment ratio of $3.80\times$ at $k = 64$ and $1.81\times$ at $k = 256$, indicating that the RL and SFT fine-tuning endpoints occupy partially overlapping but geometrically distinct regions of the base model’s principal subspace.

The RL-SFT gap is statistically robust across the full architecture. A Wilcoxon signed-rank test on per-matrix $r(k)$ values yields $p < 1.3 \times 10^{-32}$ at every k tested; 99% of the 196 weight matrices show $r_{\text{RL}} > r_{\text{SFT}}$ (Figure 3).

The per-matrix distribution is right-skewed ($r_{\text{RL}} = 2.78 \pm 2.57$, $r_{\text{SFT}} = 1.47 \pm 2.19$ at $k = 256$; mean $\pm$ std), with a small number of outlier layers reaching $r > 10$. Only 2–3

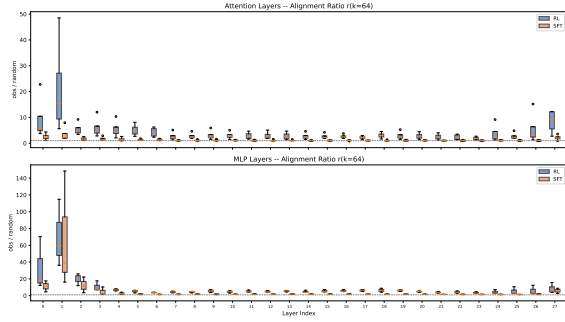


Figure 3: Per-layer SVD alignment ratio $r(k=256)$ for RL-base and SFT-base deltas across all 196 weight matrices. RL alignment exceeds SFT in 99% of layers (Wilcoxon $p < 1.3 \times 10^{-32}$). Boxes show IQR; whiskers extend to $1.5 \times$ IQR.

layers show $r_{\text{SFT}} > r_{\text{RL}}$ (most notably layer 1 `mlp.down.proj`), and in these cases both ratios remain well above random ($r > 2$), indicating that the exception layers are high-alignment for *both* methods rather than SFT-dominant.

This subspace gap is *consistent with* the observed distillation failure, though it does not by itself establish causation (§3 tests this directly). The RL policy’s weight changes are tightly concentrated along the base model’s most important directions, while SFT, trained on the *same* RL trajectories, distributes its changes more diffusely, landing closer to the random baseline.

Layer-type decomposition. The alignment gap between RL and SFT is not uniform across layer types. At $k=256$, MLP layers show stronger RL-base alignment ($3.49\times$) than attention layers ($2.25\times$), while SFT-base alignment is $2.01\times$ for MLP and $1.07\times$ for attention. The attention result is striking: SFT attention weight changes are nearly indistinguishable from random ($1.07\times$), while RL attention changes are $2.25\times$ supra-random. This suggests that the multi-objective capability encoded in attention weights is particularly resistant to SFT transfer.

Methodological considerations. Both the RL and SFT models compared here are full-rank fine-tunes of the same Qwen2.5-7B base, ensuring that the alignment ratios reflect differences in training objective rather than architectural or rank confounds. The random baseline ($n=100$ full-rank Gaussian matrices per weight, matched in shape and variance) is consistent with both models’ training regime.

U-side / V-side asymmetry. The projection decomposition reveals a structural difference between attention and MLP layers. For attention layers, the U-side (output direction) alignment is far stronger than the V-side: 0.31 vs. 0.12 at $k=256$. RL modifies how attention writes its output much more than how it reads its input. For MLP layers, the pattern reverses: U-side = 0.05 , V-side = 0.11 . RL modifies how MLP layers select their input features more than how they project outputs.

This asymmetry adds a further dimension to the alignment gap: a successful distillation method would need to replicate not just the magnitude of RL weight changes but their directional structure—which side of the SVD decomposition to modify and by how much—across different layer types. The gap at the aggregate level ($1.47\times$ vs. $2.78\times$) suggests that behavior cloning does not recover this structure, a hypothesis we test causally in §3.

2.3 Phase 1c: Activation Probing (Negative Result)

A natural follow-up is whether the RL capability is recoverable from activations, enabling “probe-then-steer” interventions (Li et al., 2023) that bypass weight-space alignment. We test this with both linear and nonlinear probes on activation differences $\Delta \mathbf{h}_l = \mathbf{h}_l^{\text{RL}} - \mathbf{h}_l^{\text{base}}$ ($n=300$ prompts, 6 layers, $k \in \{64, 256\}$). Neither succeeds: linear probes achieve binary accuracy of 0.72 , below the majority-class baseline of 0.76 , with negative R^2 for regression. MLP probes (2-layer, hidden dim 128) perform *worse* than linear probes in 47 of 55 feature/layer combinations ($p = 4.83 \times 10^{-11}$, paired t -test). A random-feature noise floor analysis confirms that 94.5% of real probe $|\rho|$ values fall within the range achievable by random projections. Full details are in Appendix C.

Within the probe families tested (single-layer linear and 2-layer MLP with hidden dimension 128), we find no recoverable activation-space signal corresponding to the weight-space SVD signature. The question of where the bottleneck ultimately resides is taken up in §3 and §5.

3 Distillation Attempt and Failure

Analysis

The SVD analysis (§2.2) reveals a geometric gap between RL and SFT weight changes, but geometry alone does not establish a causal link to distillation failure. In this section, we test whether preference-based learning (DPO) can close the gap, uncover a truncation confound that initially masked the true effect size, and examine behavioral evidence for why the student fails.

3.1 Experimental Design

Training pipeline. We train a pooled SFT baseline and three per- N DPO variants on Qwen2.5-3B (LoRA (Hu et al., 2022), rank 32). The SFT stage uses 1,118 partial-credit trajectories (pooled across $N \in \{2, 4, 8\}$; inclusion criterion: $\text{total_em} \geq 1$) for 3 epochs with $\text{lr} = 2 \times 10^{-4}$. Each DPO stage initializes from the SFT checkpoint and trains on N -specific preference pairs matched at 176 raw pairs (≈ 109 –114 train after hash-based split), with $\beta = 0.05$, $\text{lr} = 2 \times 10^{-5}$, and 1 epoch. Training details and convergence curves are in Appendix A.

Matched-sample design. The matched-sample constraint is critical for fair cross- N comparison but severely limits data volume. At $N = 4$, the teacher produces only 176 usable preference pairs (from 500 prompts at 15.2% perfect rate); at $N = 8$, only 0.4% of trajectories achieve perfect scores. We subsample all DPO conditions to the $N = 4$ bottleneck of 176 raw pairs. This scarcity is a *structural consequence* of multi-objective collapse: the teacher’s own declining success rate at high N limits the supply of contrastive training signal, and generating more prompts does not resolve it.

Evaluation protocol. All models are evaluated on held-out test sets (hash-based split, $\text{test_frac} = 0.25$) for $N \in \{2, 4, 8\}$ with $\text{max_new_tokens} = 4096$, yielding 251, 131, and 125 test prompts respectively. We report em_partial (primary metric; per-objective EM averaged across objectives and prompts), em_full (all- N -correct), and truncated_rate (fraction of outputs reaching the token limit). Statistical significance is assessed via paired bootstrap (1,000 resamples). We pre-registered a practical-significance threshold of $\Delta \text{em_partial} \geq +5\text{pp}$ at $N = 8$ with $p < 0.05$ before Stage 1 training began, representing

Model	$N = 2$	$N = 4$	$N = 8$
SFT shared (3B)	0.112	0.084	0.025
SFT full-rank (7B) [‡]	0.171	0.147	0.079
DPO $N = 2$	0.128	0.095	0.031
DPO $N = 4$	0.128	0.099	0.026
DPO $N = 8$	0.130	0.082	0.035
RL teacher (7B)	0.59	0.54	0.455

Table 2: em_partial across distillation methods and evaluation conditions ($\text{max_new_tokens} = 4096$).

[‡]The 7B full-rank SFT uses pooled training on $N \in \{2, 4, 8\}$ (same data and inclusion criteria as the 3B student). The 7B–3B gap accounts for *at most* ≈ 12 –14% of the teacher–student performance gap, remarkably stable across N (§3.2). DPO variants show marginal gains; best $N = 8$: $\text{DPO}_{N=8}$ at +1.0pp ($p = 0.163$).

$\approx 11\%$ recovery of teacher capability—the minimum we deemed meaningful for justifying the DPO pipeline.

3.2 DPO vs. SFT: Matched-Sample Comparison

Table 2 reports the full evaluation matrix. Under fair comparison (both SFT and DPO evaluated with $\text{max_new_tokens} = 4096$), DPO provides marginal gains that do not reach practical significance.

Capacity isolation. To isolate the capacity contribution, we train a 7B full-rank SFT on the same pooled data ($N \in \{2, 4, 8\}$, identical inclusion criteria) used for the 3B LoRA student. The 7B model consistently outperforms the 3B student: 0.171 vs. 0.112 at $N = 2$, 0.147 vs. 0.084 at $N = 4$, and 0.079 vs. 0.025 at $N = 8$. Defining the capacity contribution as $(\text{em}_{7\text{B}} - \text{em}_{3\text{B}})/(\text{em}_{\text{teacher}} - \text{em}_{3\text{B}})$, the 7B–3B gap accounts for *at most* 12.3%, 13.8%, and 12.6% of the teacher–student gap at $N = 2, 4$, and 8 respectively—a remarkably stable ≈ 12 –14% upper bound across objective counts. This is an upper bound because the 7B model uses full-rank fine-tuning (more expressive than LoRA), so any capacity advantage is conflated with the training-method advantage. The 7B model also substantially reduces truncation (16.7% vs. 30.7% at $N = 2$; 60.8% vs. 83.2% at $N = 8$), indicating that the larger model better controls output length. Nevertheless, the 7B SFT’s best performance (0.171) remains at 29% of the RL teacher (0.59), confirming that capacity is a minor con-

tributing factor while the SFT method itself—specifically, behavioral mismatch (§3.5)—is the dominant bottleneck. We note that the 7B model uses full-rank fine-tuning while the 3B student uses LoRA (rank 32), so this comparison is a best-effort capacity isolation (§5.1).

The best result—DPO_{N=8} at eval $N = 8$ —yields $\Delta = +1.0\text{pp}$ ($p = 0.163$, 95% CI $[-0.9, +2.9]\text{pp}$), far below the pre-registered practical-significance threshold of $+5\text{pp}$. A post-hoc power analysis confirms this is an informative null: with $n = 125$ paired test prompts and observed $\sigma_{\text{diff}} \approx 10.8\text{pp}$, we achieve > 0.99 power to detect a 5pp effect and 0.80 power at 2.7pp . The non-significant result is not due to insufficient power but reflects a genuinely small effect. We note that the data scarcity is structural rather than remediable (§5.1).

No systematic N -matching effect is observed: training on $N = k$ preference pairs does not specifically benefit evaluation at $N = k$. DPO_{N=2} performs comparably to DPO_{N=8} across all conditions, suggesting that the preference signal—regardless of objective count—provides similarly limited benefit.

Scale of failure. Both students retain $<8\%$ of teacher performance at $N = 8$: SFT at 0.025 (94.5% drop) and DPO_{N=8} at 0.035 (92.3% drop). Neither achieves $\text{em_full} > 0$ at $N \geq 4$.

3.3 Truncation Confound

An important methodological episode shaped our analysis. An earlier evaluation used $\text{max_new_tokens} = 2048$, under which the DPO advantage appeared significant (Table 3). Re-evaluating SFT with the same 4096-token budget used for DPO revealed that the original significance was largely an artifact.

At 2048 tokens, 93.6% of SFT outputs were truncated before the `<answer>` tag; extending to 4096 allowed $\sim 10\%$ more to complete, lifting em_partial from 0.011 to 0.025 . DPO’s score was unchanged at 0.035 —its apparent advantage came from SFT being disproportionately penalized. This confound is specific to RL-distilled models whose procedural strategies produce verbose outputs; we report it as a methodological caution for future work.

	em_partial	trunc. rate	Δ vs SFT	p
<i>SFT at max_new_tokens = 2048 (confounded):</i>				
SFT shared	0.011	93.6%	—	—
DPO $N=8$	0.035	— [†]	$+2.4\text{pp}$	< 0.001
<i>SFT at max_new_tokens = 4096 (corrected):</i>				
SFT shared	0.025	83.2%	—	—
DPO $N=8$	0.035	$\sim 84\%$	$+1.0\text{pp}$	0.163

Table 3: Truncation confound at $N = 8$. The initial SFT evaluation at 2048 tokens truncated 93.6% of outputs, yielding a deceptively significant DPO advantage. Extending SFT’s budget to 4096 tokens reduced truncation to 83.2%, lifted em_partial from 0.011 to 0.025 , and revealed that the DPO advantage is not significant. [†]DPO was evaluated at $\text{max_new_tokens} = 4096$ in both conditions; only the SFT budget differed.

3.4 Behavioral Mismatch

The truncation confound points to a deeper issue: why are the student’s outputs so long that 83% hit the 4096-token limit at $N = 8$? The answer reveals a behavioral mismatch between what the student learns and what it can execute.

Observational evidence. The evidence in this subsection is *observational*, not causal (§5.1). The SFT student reproduces the teacher’s multi-turn action structure (`<think>` $\rightarrow$ `<search>` $\rightarrow$ `<information>` $\rightarrow$ `<answer>`), but without the retrieval API, search and information segments contain hallucinated content (50–200 tokens per block). At $N = 8$, these hallucinated search cycles consume the majority of the 4096-token budget before the first `<answer>` tag.

Truncation scaling with N . Truncation scales monotonically: 30.7% at $N = 2$, 55.7% at $N = 4$, 83.2% at $N = 8$ —consistent with generation length growing proportionally to inherited search procedures.

Critically, DPO does not alleviate this pattern. The DPO models exhibit comparable truncation rates ($\sim 84\%$ at $N = 8$), confirming that preference learning does not teach the student to adopt a more token-efficient strategy—it inherits the same procedural behavior as SFT.¹

¹A pilot attempt to suppress search tokens via bigram blocking was inconclusive: the model generated tag variants (e.g., malformed closing tags, inserted whitespace) that bypassed token-level suppression, suggesting that the behavioral pattern is encoded at a level above individual token sequences.

	$N=2$	$N=4$	$N=8$
SFT (no API)	0.112	0.084	0.025
SFT + oracle API	0.530	0.521	0.498
Δ	+41.8pp	+43.7pp	+47.3pp
p (bootstrap)	<0.0001	<0.0001	<0.0001
RL teacher [†]	0.59	0.54	0.455

Table 4: Effect of providing oracle retrieval results at inference time. With real search results injected, the SFT student recovers from near-zero to near-teacher performance. Paired bootstrap, 10,000 resamples. [†]Teacher evaluated on a different prompt set; direct comparison is not valid.

3.5 Causal Verification: Oracle Retrieval

The observational evidence in §3.4 raises a natural intervention: if the student’s failure stems from executing search procedures without a retrieval backend, then *providing* the backend should recover performance. We test this by intercepting the student’s `<search>` outputs at inference time and injecting oracle retrieval results from the RL teacher’s trajectories (the same passages the teacher used for each test prompt).

Table 4 shows that providing real retrieval results transforms student performance: `em_partial` jumps from 0.025 to 0.498 at $N=8$ (+47.3pp, $p < 0.0001$, 95% CI [43.8, 50.8]pp). Truncation drops from 83.2% to 0%—the student completes every output when search queries return real passages instead of hallucinated content. This constitutes causal evidence that behavioral mismatch, not capacity limitations alone, drives the distillation failure: the student has learned a viable search strategy but lacks the execution environment to realize it.

Realistic retrieval: BM25. The oracle setting uses teacher passages and thus provides an upper bound. To assess a deployable configuration, we replace oracle retrieval with BM25 (top-4 passages from the same corpus). Table 5 and Figure 4 show that BM25 retrieval recovers a substantial fraction of the oracle gain: `em_partial` improves from 0.025 to 0.147 at $N=8$ (+12.2pp), with truncation dropping to 0% across all N . The BM25 student recovers 14–26% of the oracle improvement, with the largest relative gain at $N=8$ where the no-API baseline is most degraded. The gap between BM25 and oracle (0.147 vs. 0.498 at $N=8$) reflects retrieval quality rather than student capability: the stu-

	$N=2$	$N=4$	$N=8$
SFT (no API)	0.112	0.084	0.025
SFT + BM25	0.203	0.147	0.147
SFT + oracle API	0.530	0.521	0.498
RL teacher	0.59	0.54	0.455

Table 5: `em_partial` under different retrieval conditions. BM25 retrieval recovers a meaningful fraction of the oracle gain (+9.2pp at $N=2$, +12.2pp at $N=8$) and eliminates truncation entirely (0% across all N). The remaining gap to oracle reflects retrieval quality, not student strategy.

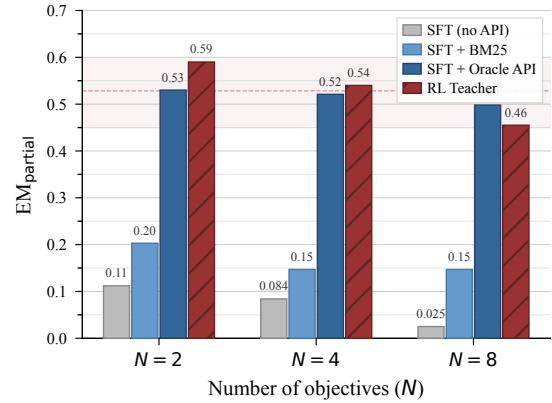


Figure 4: `em_partial` under different retrieval conditions. Providing oracle retrieval results at inference time recovers the SFT student from near-zero to near-teacher performance across all N , while even a standard BM25 backend yields meaningful gains. The RL teacher (hatched) is shown for reference.

dent’s queries closely match the teacher’s (Jaccard = 0.95), but BM25’s top-4 recall is substantially lower than oracle injection.

Query fidelity and search behavior. The student averages exactly 1.0 search per prompt across all 507 test cases. Crucially, this is not a simplification of the teacher’s strategy: analysis of the teacher’s training trajectories reveals that >95% of teacher responses also contain a single search round (mean searches: 1.04 at $N=2$, 1.01 at $N=4$, 1.06 at $N=8$), because the prompt prefix already includes the first retrieval result. Furthermore, the student’s search queries closely match the teacher’s: mean Jaccard similarity is 0.95 (median = 1.0, $n=507$). SFT thus faithfully transfers both the search pattern and the query formulation—the bottleneck is purely the absence of a retrieval backend, not a failure of strategy learning.

Why `em_full` remains low. Despite recovering `em_partial` to near-teacher levels, `em_full` remains low under oracle retrieval (0.008 at $N=8$). Per-objective analysis reveals that individual objective accuracy is $\approx 50\%$ (range 38–58% across positions), with no systematic decay across objective index. The low `em_full` follows from probability stacking: if each of N objectives is independently correct with probability ≈ 0.5 , the probability of *all* N being correct is $\approx 0.5^N$ (0.4% at $N=8$), consistent with the observed 0.8%. The single search round retrieves information relevant to a subset of objectives; full coverage would require multiple targeted searches.

3.6 Synthesis

The analyses in this section converge on a consistent diagnostic:

- 1. Preference learning does not close the geometric gap.** RL weight changes are $2\text{--}6\times$ supra-random aligned with the base model’s principal directions (§2.2), while SFT achieves only $1.47\times$. DPO—which provides richer training signal—does not bridge this gap: all DPO variants remain within 1pp of SFT at $N=8$.
- 2. The student imitates a strategy it cannot execute—causally supported.** The RL teacher’s multi-objective capability is inseparable from its iterative search procedure. SFT transfers this procedure, but without the retrieval backend, hallucinated search traces exhaust the generation budget. Providing oracle retrieval at inference time recovers `em_partial` from 0.025 to 0.498 at $N=8$ ($+47.3\text{pp}$, $p < 0.0001$; §3.5), confirming that the student possesses a viable search strategy but lacks execution capability.
- 3. Truncation masks and distorts the comparison.** At insufficient generation budgets, truncation disproportionately penalizes the more verbose model (SFT), creating a spurious advantage for DPO. After controlling for this confound, the DPO advantage shrinks from $+2.4\text{pp}$ ($p < 0.001$) to $+1.0\text{pp}$ ($p = 0.163$)—both methods fail for the same reason.

The oracle retrieval experiment (§3.5) elevates behavioral mismatch from observational hypothesis to a causally supported finding, and the BM25 experiment confirms that even a standard retrieval backend recovers meaningful performance ($+12.2\text{pp}$ at $N=8$). Notably, the

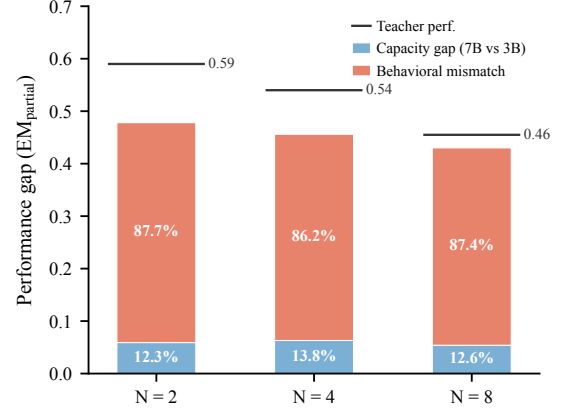


Figure 5: Decomposition of the teacher–student performance gap across objective counts. Capacity (7B vs. 3B SFT) accounts for a stable $\sim 12\text{--}14\%$, while behavioral mismatch dominates at $\sim 86\text{--}88\%$.

student faithfully reproduces the teacher’s search strategy (Jaccard = 0.95; §3.5), confirming that the bottleneck is execution capability rather than strategy learning. A 7B pooled SFT (§3.2) quantifies the capacity contribution at *at most* $\approx 12\text{--}14\%$ of the teacher–student gap—a minor but consistent factor across all N values, while behavioral mismatch accounts for the remaining $\approx 86\text{--}88\%$ (Figure 5).

We note that capacity and behavioral mismatch may interact non-additively, and additional factors (e.g., training-method differences) may also contribute; these percentages should be interpreted as approximate decompositions rather than exact attributions.

4 Related Work

RL-trained LLM agents. RL fine-tuning endows language models with procedural capabilities that SFT does not produce. MEM1 (Zhou et al., 2025), WebArena (Zhou et al., 2024a), AgentBench (Liu et al., 2024), ReAct (Yao et al., 2023), and AgentTuning (Zeng et al., 2024) all demonstrate multi-step action strategies for tasks ranging from QA to web navigation. We investigate why these RL-acquired strategies resist distillation.

Knowledge distillation for LLMs. Standard methods (Hsieh et al., 2023; Gu et al., 2024; Jiang et al., 2023) target single-turn transfer on static inputs. Policy distillation in RL (Rusu et al., 2016; Czarnecki et al., 2019) and interactive imitation learning (Dagger; Ross et al.,

2011) transfer procedural knowledge but in low-dimensional action spaces. Our setting differs: the teacher’s competence arises from a *multi-turn procedural policy* in the high-dimensional token space of LLMs.

SFT–RL alignment gaps and weight-space analysis. The “alignment tax” (Askell et al., 2021) documents RLHF degrading pre-training capabilities. Jin et al. (2025a) show SFT induces “hard alignment” while RL produces “soft alignment”; Aghajanyan et al. (2021) demonstrate fine-tuning operates in low intrinsic dimension. PiSSA (Meng et al., 2024) initializes LoRA from principal singular components of the pre-trained weight matrix; our SVD alignment ratio measures a related but distinct quantity—the overlap of *fine-tuning deltas* with the base model’s principal subspace, rather than using that subspace for parameter-efficient initialization. Loss landscape visualization (Li et al., 2018; Fort and Ganguli, 2019) provides complementary geometric perspectives on fine-tuning. We extend these by revealing quantitative gaps ($2\text{--}6\times$ for RL vs. $1.5\times$ for SFT) not captured by prior metrics.

Multi-objective RL for LLMs. Recent work explores multi-objective optimization in LLM alignment: Rewarded Soups (Rame et al., 2024) interpolates weights fine-tuned on diverse rewards to approximate the Pareto front, MODPO (Zhou et al., 2024b) extends DPO to handle multiple preference dimensions, and fine-grained human feedback (Wu et al., 2024) decomposes rewards into per-aspect signals. These methods address *training* with multiple objectives; our work addresses the complementary problem of *distilling* multi-objective RL policies, where the challenge is not balancing objectives during training but transferring the teacher’s coordinated multi-objective execution strategy.

Preference learning and DPO. DPO (Rafailov et al., 2023) and extensions (IPO (Gheshlaghi Azar et al., 2024), KTO (Ethayarajh et al., 2024), RAFT++ (Dong et al., 2023)) provide offline alternatives to RLHF. We use DPO as a stronger-than-SFT baseline; its failure ($+1.0\text{pp}$, not significant) suggests behavioral mismatch rather than insufficient signal richness. Following Karl et al. (2024), we frame this as a diagnostic contribution.

5 Conclusion

We have presented a three-phase diagnostic framework for understanding why SFT distillation of multi-objective RL policies fails catastrophically. Applied to MEM1—a 7B RL agent for retrieval-augmented QA—our analysis yields three findings:

1. Multi-objective performance degrades monotonically with objective count N (EM $0.62 \rightarrow 0.341$ from $N = 1$ to $N = 16$), with no evidence of a phase transition within the tested range. The distillation target is to slow this decline, not to recover from a cliff.
2. RL weight changes concentrate $2\text{--}6\times$ supranormal along the base model’s principal singular directions, while SFT distillation recovers significantly less of this geometric structure ($1.47\times$ vs. $2.78\times$ at $k = 256$; Wilcoxon $p < 1.3 \times 10^{-32}$ across 196 weight matrices, with 99% of layers showing $r_{\text{RL}} > r_{\text{SFT}}$).
3. Matched-sample DPO—a stronger training signal than SFT—fails to close the gap: the best variant improves over SFT by only $+1.0\text{pp}$ at $N = 8$ ($p = 0.163$), far below practical significance. A truncation confound analysis reveals that an earlier apparent advantage ($+2.4\text{pp}$, $p < 0.001$) was partly an artifact of insufficient generation budget penalizing SFT disproportionately.

Taken together, these results indicate that the distillation bottleneck for multi-objective RL policies is not addressable by richer preference signals alone. A causal intervention—providing oracle retrieval results at inference time—recovers $\text{em}_{\text{partial}}$ from 0.025 to 0.498 at $N = 8$ ($+47.3\text{pp}$, $p < 0.0001$), confirming that behavioral mismatch is a primary driver: the student learns a viable search strategy but cannot execute it without the retrieval backend. The student faithfully reproduces the teacher’s search behavior (mean query Jaccard = 0.95, single-round pattern matching the teacher’s $>95\%$ single-search training data), indicating that SFT successfully transfers procedural knowledge—the failure is purely in execution capability. From a practical standpoint, our results suggest that equipping the SFT student with a retrieval API at inference time—even a simple BM25 system—can substantially recover performance ($+12.2\text{pp}$ at $N = 8$), offering an immediately deployable mitigation that requires no additional training.

769 5.1 Limitations

770 **Scope.** All experiments use one RL teacher
771 (MEM1, 7B) on one domain (multi-objective
772 RAG QA). A 7B full-rank pooled SFT quanti-
773 fies the capacity contribution at $\approx 12\text{--}14\%$ of the
774 teacher–student gap across all N (§3.2); how-
775 ever, the training-method mismatch (full-rank vs.
776 LoRA) makes this best-effort isolation. The SVD
777 Gaussian random baseline is validated by the
778 Wilcoxon test ($p < 1.3 \times 10^{-32}$, $n = 196$).

779 **DPO power and future work.** DPO uses only
780 176 matched pairs (0.4% perfect rate at $N = 8$),
781 but power > 0.99 for 5pp (floor 2.7pp). Ora-
782 cle retrieval uses teacher passages (upper bound);
783 BM25 retrieval recovers +9–12pp (§3.5), con-
784 firming the finding generalizes beyond the oracle
785 setting. Training a 3B RL baseline and cross-task
786 generalization remain as future work.

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966 A Training Details

967 Table 6 summarizes the hyperparameters for all
968 training stages.

969 **SFT convergence (3B).** The 3B SFT stage
970 converges within 3 epochs with monotonically
971 decreasing training loss (final loss ≈ 0.91). No
972 overfitting is observed on the held-out split.

973 **7B capacity-isolation model.** Table 7 summa-
974 rizes the hyperparameters for the 7B full-rank
975 pooled SFT used in the capacity-isolation exper-
976 iment (§3.2).

977 **DPO convergence.** All three DPO variants
978 ($N \in \{2, 4, 8\}$) train for 1 epoch. Final train-
979 ing losses range from 0.595 (DPO _{$N=2$}) to 0.678
980 (DPO _{$N=8$}). The rewards/margins are positive
981 but small: 0.207 (DPO _{$N=2$}), 0.100 (DPO _{$N=4$}),
982 0.056 (DPO _{$N=8$}), consistent with the limited
983 preference signal available from 176 matched
984 pairs. The decreasing margin with N reflects the
985 increasing difficulty of producing contrastive sig-
986 nal at higher objective counts.

	SFT (pooled)	DPO (per- N)
Base model	Qwen2.5-3B	SFT checkpoint
LoRA rank	32	32
LoRA α	64	64
LoRA target	all linear	all linear
Learning rate	2×10^{-4}	2×10^{-5}
Epochs	3	1
Batch size	4	4
Gradient accum.	4	4
Max seq. length	6144	6144
Optimizer	AdamW	AdamW
Weight decay	0.01	0.01
Warmup ratio	0.03	0.1
DPO β	—	0.05
Training samples	1,118 (partial-credit, pooled $N \in \{2, 4, 8\}$)	176 raw pairs (matched to $N = 4$ bottleneck)
Train/test split	hash-based, test_frac = 0.25	

Table 6: Training hyperparameters for SFT and DPO stages. Both stages use LoRA (rank 32) on Qwen2.5-3B. The SFT stage uses partial-credit trajectories (inclusion: total_lem ≥ 1); DPO stages use matched-sample preference pairs subsampled to 176 raw pairs.

	7B SFT (pooled)
Base model	Qwen2.5-7B
Fine-tuning	Full-rank (no LoRA)
Learning rate	2×10^{-5}
Epochs	2
Batch size	1
Gradient accum.	4 (eff. batch 4)
Max seq. length	6144
Optimizer	AdamW
Precision	bf16
Gradient checkpointing	Yes
Training samples	1,118 (same partial-credit pooled data as 3B SFT)
Training time	19.5 min (560 steps)
Final train loss	0.83

Table 7: Training hyperparameters for the 7B full-rank capacity-isolation SFT. This model uses the same pooled data and inclusion criteria as the 3B LoRA SFT, differing only in model size (7B vs. 3B) and fine-tuning method (full-rank vs. LoRA rank 32). The lower learning rate (2×10^{-5} vs. 2×10^{-4}) and fewer epochs (2 vs. 3) reflect standard practice for full-rank fine-tuning of larger models.

B SVD Alignment Per-Layer Distribution

See Figure 3 in the main text for the per-layer SVD alignment boxplot.

C Activation Probing Details

We extract activation differences $\Delta \mathbf{h}_l = \mathbf{h}_l^{\text{RL}} - \mathbf{h}_l^{\text{base}}$ at layers $l \in \{0, 3, 7, 11, 15, 19\}$ for $n = 300$ prompts, projecting onto the top- k principal directions ($k \in \{64, 256\}$) of the base model’s weight matrices. Eight feature variants are tested: raw activations $\mathbf{h}^{\text{RL}}$, $\mathbf{h}^{\text{base}}$, their difference $\Delta \mathbf{h}$, and top- k projections thereof.

Linear probes. Binary classification (per-objective EM above/below median) achieves accuracy 0.72, below the majority-class baseline 0.76. Regression probes yield negative R^2 across all 55 (feature, layer, k) combinations.

MLP probes. Two-layer MLPs (hidden dimension 128, ReLU) trained on the same features predict per-objective EM as a continuous target (Spearman ρ). MLP probes lose to linear probes in 47/55 cases (85.5%; paired $t = -8.19$, $p = 4.83 \times 10^{-11}$, Cohen’s $d = -1.105$). Mean ρ : -0.011 (MLP) vs. 0.073 (linear).

Random-feature baseline. We construct 5 random orthonormal bases $\times 6$ layers $\times 2$ k values. The maximum $|\rho|$ under random features is 0.224; 52/55 (94.5%) of real linear probe $|\rho|$ values fall within this noise band. The per-objective

signal in $\Delta \mathbf{h}$ is indistinguishable from random projections.